

MPEG: A Video Compression Standard for Multimedia Applications

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Outline

- Introduction
- MPEG Goals
- MPEG Details
- Performance and Such
- Summary



Introduction

- 1980's technology made possible full-motion video over networks
 - Television and Computer Video seen moving closer
 - (Today, Sony and Microsoft are squaring off)
- Needed a standard
 - Often trigger needed volume production
 - *Ala facsimile* (fax)
 - Avoid *de facto* standard by industry
- 1988, Established the Motion Picture Experts Group (MPEG)
 - Worked towards MPEG-1
 - Primarily video but includes audio (MP3)

The Need for Video Compression

- High-Definition Television (HDTV)
 - 1920x1080
 - 30 frames per second (full motion)
 - 8 bits for each three primary colors (RGB)
→ Total 1.5 Gb/sec!
- Each cable channel is 6 MHz
 - Max data rate of 19.2 Mb/sec
 - Reduced to 18 Mb/sec w/audio + control ...
→ Compression rate must be ~ 80:1!

Compatibility Goals

- CD-ROM and DAT key storage devices
 - 1-2 Mbits/sec for 1x CD-ROM
- Two types of application videos:
 - Asymmetric (encoded once, decoded many)
 - Video games, Video on Demand
 - Symmetric (encoded once, decoded once)
 - Video phone, video mail ...
- *(How do you think the two types might influence design?)*
- Video at about 1.5 Mbits/sec
- Audio at about 64-192 kbits/channel

Requirements

- Random Access, Reverse, Fast Forward, Search
 - At any point in the stream (within $\frac{1}{2}$ second)
 - Can reduce quality somewhat during this task, if needed
- Audio/Video Synchronization
- Robustness to errors
 - Not catastrophic if some bits are lost
 - Lends itself to Internet streaming
- Coding/Decoding delay under 150ms
 - For interactive applications
- Editability
 - Modify/Replace frames

Relevant Standards

- Joint picture Experts Group (JPEG)
 - Compress still images only
- Expert Group on Visual Telephony (H.261)
 - Compress sequence of images
 - Over ISDN (64 kbits/sec)
 - Low-delay
- Other high-bandwidth “H” standards:
 - H21 (34 Mbits/sec)
 - H22 (45 Mbits/sec)

Outline

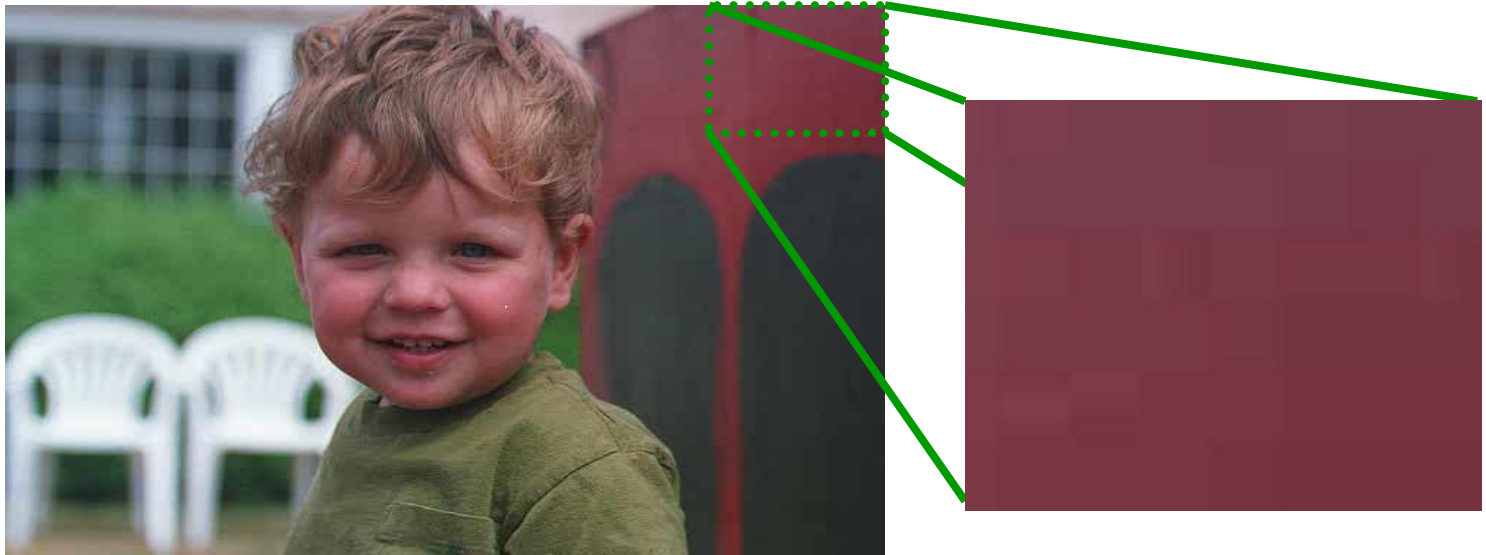
- Introduction (done)
- MPEG Goals (done)
- MPEG Details ←
- Performance and Such
- Summary

MPEG Compression

- Compression through
 - Spatial
 - Temporal

Spatial Redundancy

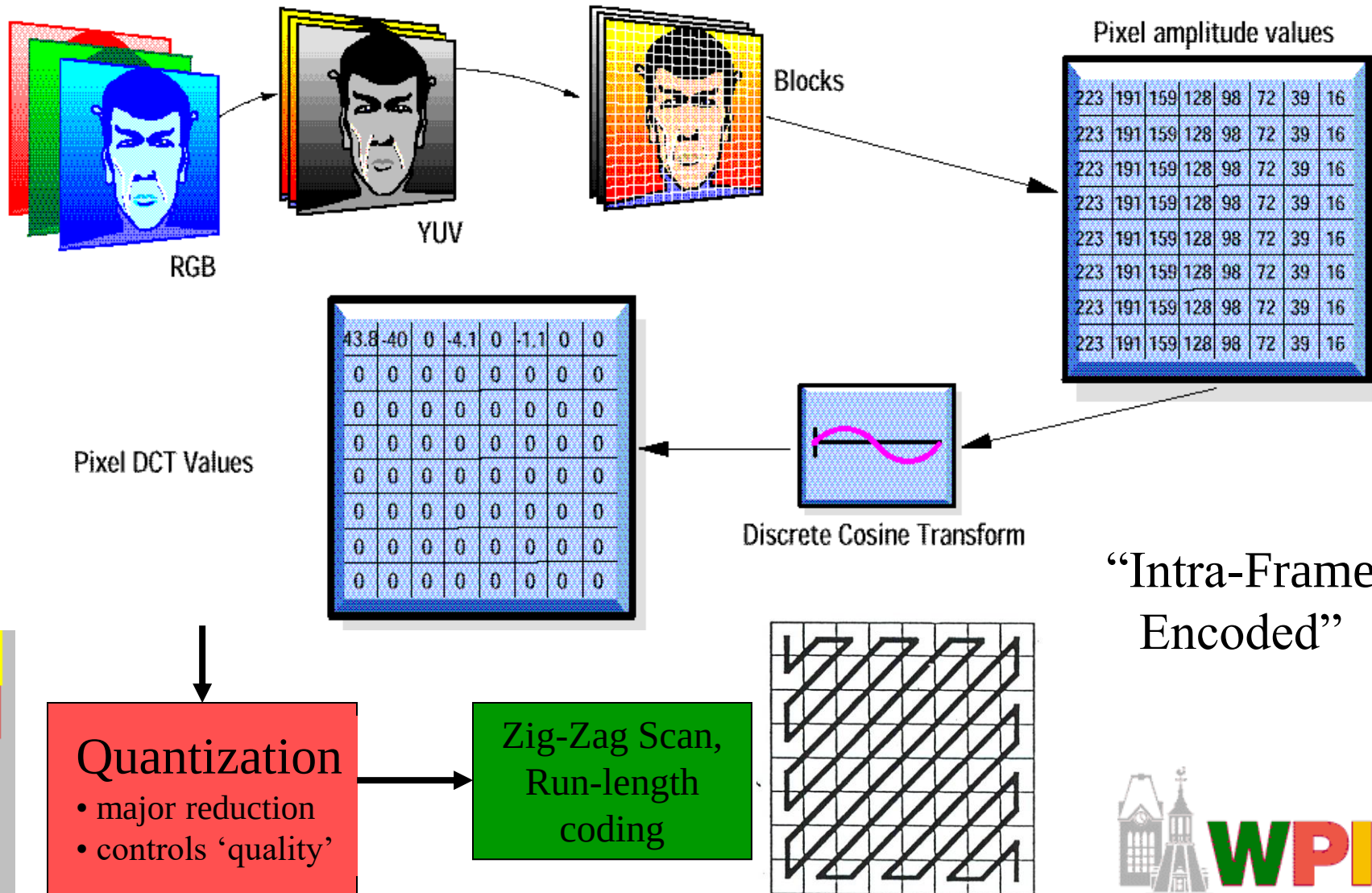
- Take advantage of similarity among most neighboring pixels



Spatial Redundancy Reduction

- RGB to YUV
 - less information required for YUV (humans less sensitive to chrominance)
- Macro Blocks
 - Take groups of pixels (16x16)
- Discrete Cosine Transformation (DCT)
 - Based on Fourier analysis where represent signal as sum of sine's and cosine's
 - Concentrates on higher-frequency values
 - Represent pixels in blocks with fewer numbers
- Quantization
 - Reduce data required for co-efficients
- Entropy coding
 - Compress

Spatial Redundancy Reduction



“Intra-Frame
Encoded”

Groupwork

- When may spatial redundancy reduction be ineffective? What kinds of images/movies?

Groupwork

- When may spatial redundancy reduction be ineffective?
 - High-resolution images and displays
 - May appear ‘coarse’
 - A varied image or ‘busy’ scene
 - Many colors, few adjacent

Loss of Resolution



← Original (63 kb)



Low (7kb)



Very Low (4 kb)



Temporal Redundancy

- Take advantage of similarity between successive frames



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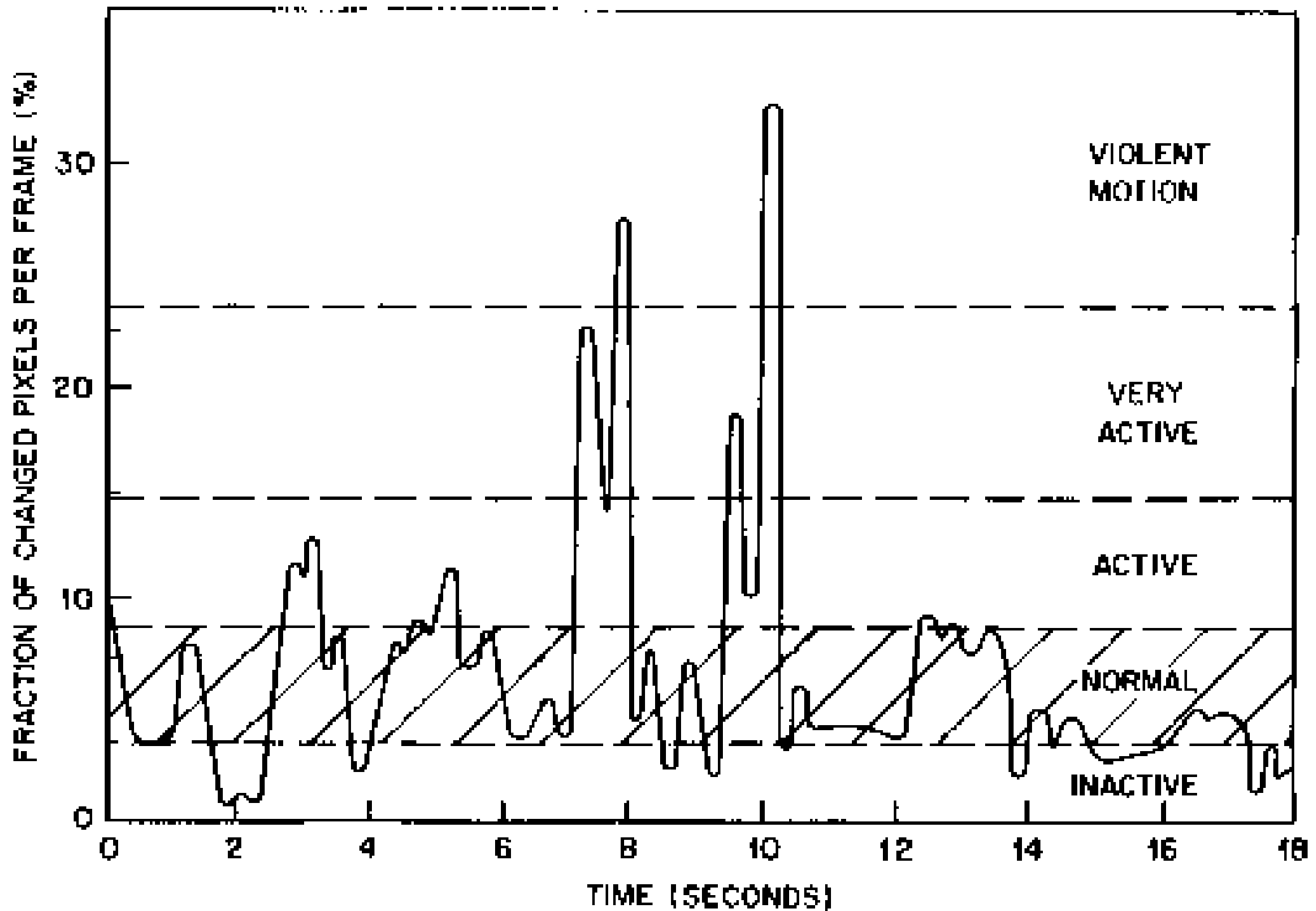


951



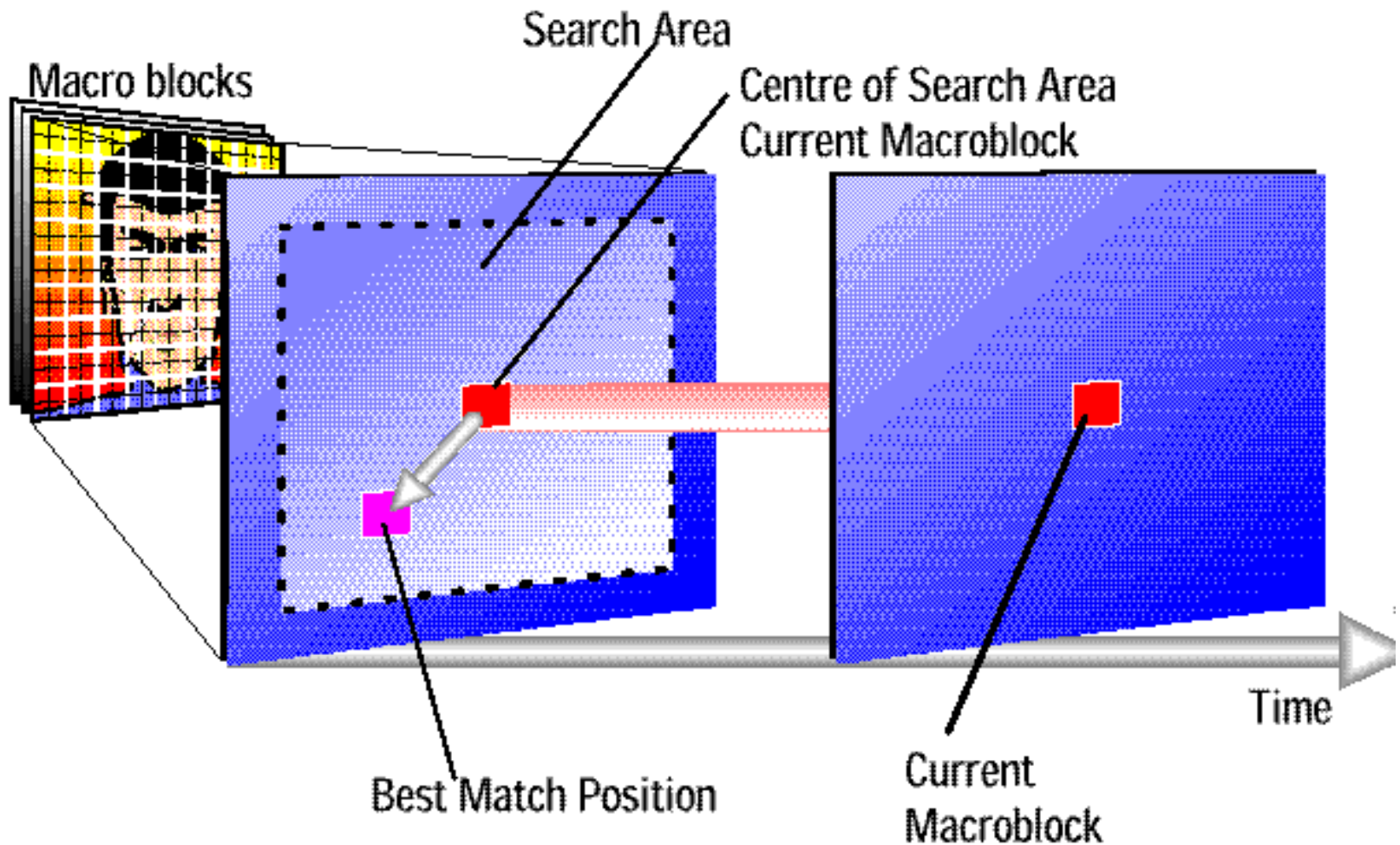
952

Temporal Activity

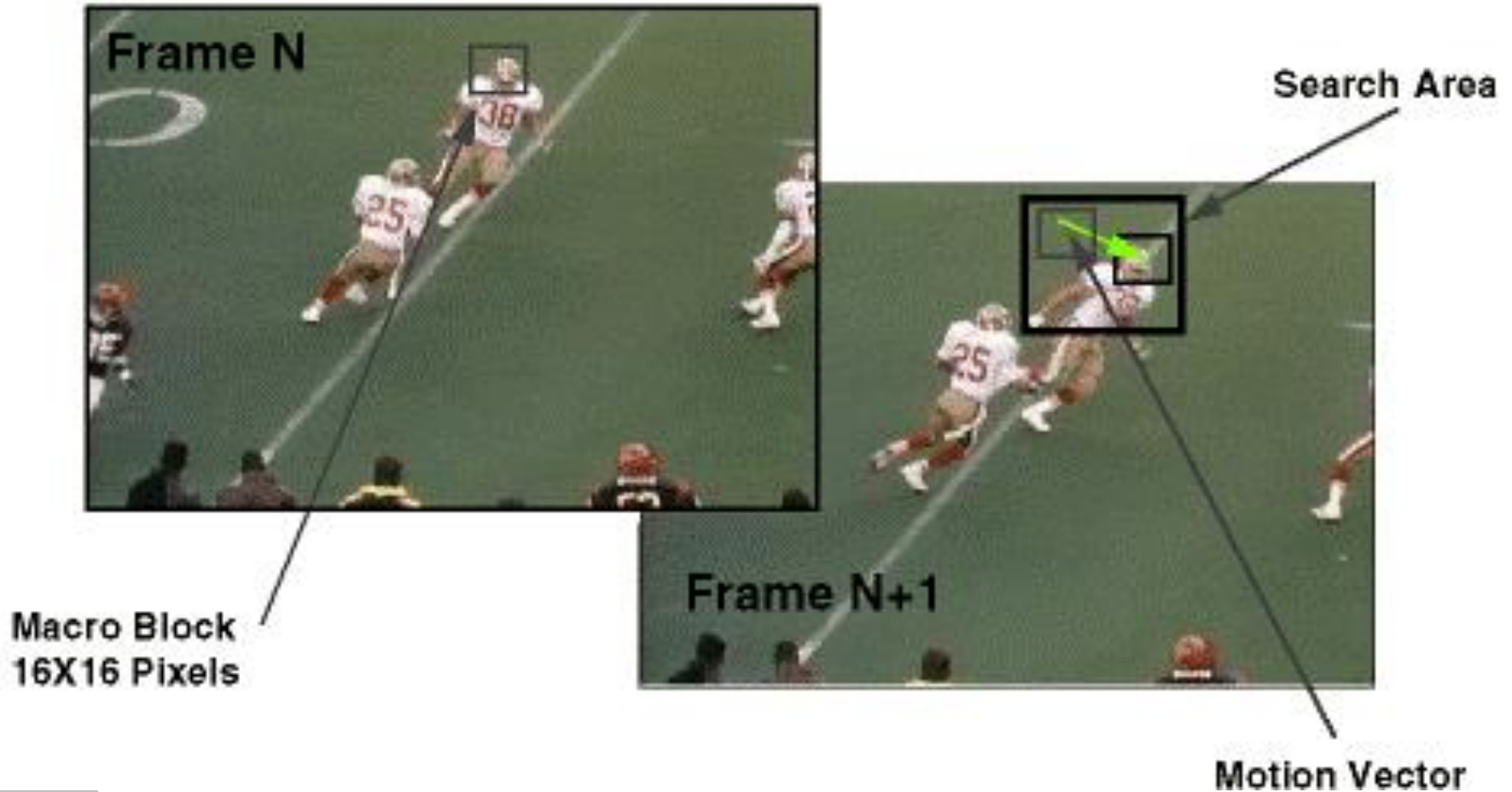


“Talking Head”

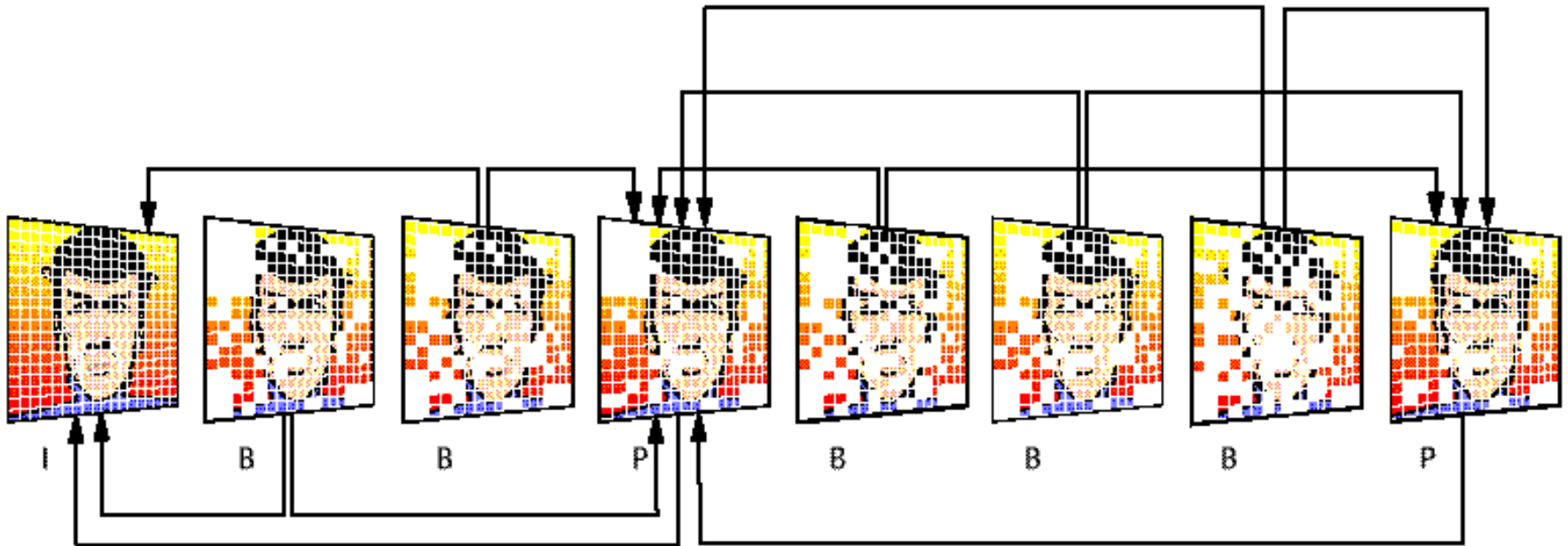
Temporal Redundancy Reduction



Temporal Redundancy Reduction



Temporal Redundancy Reduction



- *I* frames are independently encoded
- *P* frames are based on previous *I*, *P* frames
- *B* frames are based on previous and following *I* and *P* frames
 - In case something is uncovered

Group of Pictures (GOP)

- Starts with an I-frame
- Ends with frame right before next I-frame
- “Open” ends in B-frame, “Closed” in P-frame
 - (What is the difference?)
- MPEG Encoding parameter, but ‘typical’:
 - I B B P B B P B B I
 - I B B P B B P B B P B B I
- *Why not have all P and B frames?*

Groupwork

- When may temporal redundancy reduction be ineffective?

Groupwork

- *When may temporal redundancy reduction be ineffective?*
 - Many scene changes
 - High motion

Non-Temporal Redundancy

- Many scene changes vs. Few scene changes



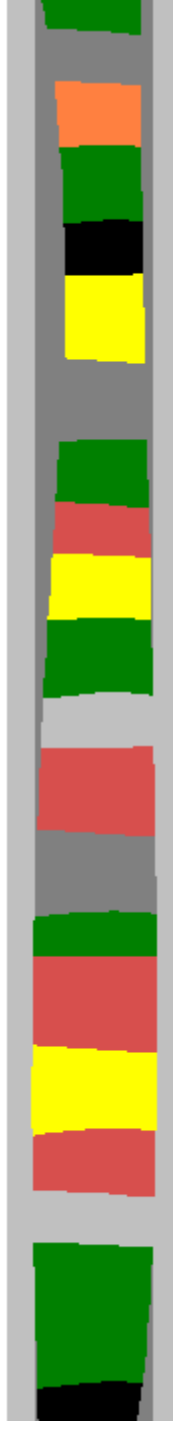
Non-Temporal Redundancy

- Sometimes high motion



Typical MPEG Parameters

Parameters	Value
Image resolution	384x288
Quantization factor	8
Frames between I pictures	5
Frames between P pictures	2
Frames sequence as to be displayed	...IBBFBI...
Rate control	None



Typical Compression Performance

Type Size Compression

I	18 KB	7:1
P	6 KB	20:1
B	2.5 KB	50:1
Avg	4.8 KB	27:1

- Note, results in Variable Bit Rate, even if frame rate is constant

MPEG Today

- MPEG video compression widely used
 - digital television set-top boxes
 - HDTV decoders
 - DVD players
 - video conferencing
 - Internet video
 - ...

MPEG Today

- MPEG-2
 - Super-set of MPEG-1
 - Rates up to 10 Mbps (720x486)
 - Can do HDTV (no MPEG-3)
- MPEG-4
 - Around *Objects*, not *Frames*
 - Lower bandwidth
 - Has some built-in repair
- MPEG-7
 - Not (yet) a standard
 - Allows content-description (ease of searching)
- MP3
 - For audio
 - MPEG Layer-3

MPEG Tools

- MPEG tools at:
 - <http://www-plateau.cs.berkeley.edu/mpeg/index.html>
- MPEG streaming at:
 - <http://www.comp.lancs.ac.uk/>